EL JOKER — Script du pitch, questions du jury et vidéo de démo

Version 2 · 18 octobre 2026 · Team Chain619 · Rihem Ghoudi & Mariem Souayh

> Rihem ouvre le pitch et mène la démo ; Mariem présente la solution et la partie technique. Total : environ 9 min 30, 15 slides. Le texte à dire est en anglais, les consignes sont en français.

Qui parle quand

EL JOKER est présenté comme **la couche de contrôle pour les agents IA qui dépensent de l'argent**, avec deux organisations (une ONG et une entreprise) et le coffre Hedera. Rihem parle environ 4 min 30, Mariem environ 4 min 40, et vous dites la dernière slide à deux.

#	Slide	Qui parle	Durée	Fin à	Message clé
1	Cover	Rihem	25 s	0:25	La couche de contrôle pour les agents IA qui dépensent
2	The question	Rihem	25 s	0:50	Qui contrôle une IA qui dépense de l'argent ?
3	The problem	Rihem	35 s	1:25	Hallucination, dépenses excessives, traces effaçables
4	One engine, any organization	Rihem	35 s	2:00	Une ONG et une entreprise : seule la politique change
5	The solution	Mariem	40 s	2:40	AI proposes. Rules decide. Hedera proves.
6	The crew	Mariem	35 s	3:15	Cinq agents, un seul utilise l'IA
7	Mission flow	Mariem	35 s	3:50	Au-dessus de la limite : le coffre Hedera
8	Demo: Approved	Rihem (live)	55 s	4:45	Un vrai paiement sur testnet, 6 preuves
9	Demo: Blocked	Mariem (Rihem lance)	50 s	5:35	Quatre refus, aucun argent ne bouge
10	Vault	Rihem (live)	55 s	6:30	Seule une signature humaine sur Hedera libère l'argent
11	Guardian	Mariem	30 s	7:00	Pas d'IA pour décider : rien à négocier
12	Four Hedera services	Mariem	45 s	7:45	HCS, transactions programmées, HTS, Mirror Node
13	Tampering + public verifier	Mariem	45 s	8:30	Hedera ne réécrit pas son passé, et chacun peut vérifier
14	Verified live	Rihem	40 s	9:10	Testé en vrai, feuille de route, limites honnêtes
15	Closing	Mariem puis Rihem	15 s	9:25	Autonomous, but never uncontrolled.

Si la démo live plante, restez sur les slides 8 à 10 et dites : « Here is the same run, recorded live on testnet earlier today. » Si vous êtes en retard à la slide 8, réduisez la slide 9 à deux exemples (budget et facture recyclée).

Avant le pitch, lancez `npm run hedera:setup -- --vault --stablecoin`, puis `npm run demo:check`. Si tout passe, dites « eleven checks passed live » à la slide 14 au lieu de « eight scenarios ». Tant que ce n'est pas fait, ne dites pas que le coffre et EJUSD ont tourné en direct sur testnet.

Script : slides 1 à 7

Le texte entre crochets est une consigne : ne le dites pas. Parlez lentement, environ 130 mots par minute, et regardez le jury, pas l'écran.

1. Cover — Rihem (25 s)

[Laissez l'image s'afficher, souriez, puis parlez.]

> « Good morning. We are team Chain619. I'm Rihem, and this is Mariem. This is EL JOKER. In Tunisian, *5aliha 3lina* means "leave it to us". EL JOKER is the control layer for AI agents that spend money. One agent uses AI to propose. Four are strict code that verify, decide and pay. And above the limit, the money sits in a Hedera vault that only a human co-signature can open. »

2. The question — Rihem (25 s)

> « AI agents are moving from talking to acting: they book, they order, they pay. The moment an agent can spend money, two questions matter. Who is in control when it is wrong, or when someone tricks it? And how can anyone prove, afterwards, what it decided? »

3. The problem — Rihem (35 s)

> « Today, agents fail in three ways. They hallucinate: they invent a supplier, a price or an approval, and act on it. They overspend: they forget the budget, or pay the same invoice twice. And they hide the trail: their logs live in a database that anyone with access can edit. So every organization has to choose: autonomy, or control. We remove that trade-off. »

4. One engine, any organization — Rihem (35 s)

> « Where will agents spend first? Everywhere money moves. We run two examples live. An NGO: a water network fails in Gabès, 500 people need water and medicine in 48 hours, with 5,000 dollars. And a company: ACME needs 12 laptops for a new team, 15,000 dollars. Same five agents, same proofs on Hedera. Only the policy changes: 5,000 dollars of autonomy for the NGO, 2,000 for the company. We start with the hardest case, the emergency. Mariem? »

5. The solution — Mariem (40 s)

> « Thank you, Rihem. Our answer fits in one line: AI proposes, rules decide, Hedera proves. Claude ranks the supplier offers and explains the trade-offs. But the AI never touches the money. An independent verifier checks its work. A guardian with fixed rules decides, with one policy per organization. Above the limit, a human must co-sign on Hedera itself before anything moves. And every step is written to the Hedera Consensus Service. »

6. The crew — Mariem (35 s)

[Montrez chaque robot de la main en le nommant.]

> « Meet the crew, with Tunisian names. Lem3alem, "the boss", turns the request into a validated mission and never guesses a missing fact. Lemhaf, "the clever one", is the only agent powered by AI: Claude. Nasnes, "the one who watches", verifies everything independently. 3ases, "the guard", applies the rules

> with no AI at all. And Kanawita, "the money", pays only when authorized. Above the limit, it can only lock the money. »

7. Mission flow — Mariem (35 s)

> « Here is one mission, end to end. The operator types the need. Lem3alem plans it. Lemhaf asks Claude to rank the offers. Nasnes runs ten independent checks. 3ases applies the policy. Above the limit, the payment is locked in a Hedera vault until a human signs. Kanawita pays exactly once, and every step leaves a receipt on Hedera. Rihem, show them. »

Script : slides 8 à 15

Si vous faites la démo en direct, l'application doit déjà être ouverte dans un autre onglet, avec la demande pré-remplie. Le temps que Claude réponde (environ 10 s), continuez à parler.

8. Demo: Approved — Rihem, en direct (55 s)

[Passez à l'application, appuyez sur Envoyer.]

> « This is a real run, on Hedera testnet, with Claude. I send the Gabès request. Claude compares three offers and picks Oasis at 4,150 dollars, and explains why. Nasnes passes ten out of ten checks. 3ases passes all twelve checks it runs here, risk low: approved. Kanawita pays and the receipt says SUCCESS. On the right, the six decisions of this mission appear on Hedera one by one, each with a HashScan link anyone can open. »

[Si EJUSD est configuré, ajoutez : « It pays 4,150.00 EJUSD, our stablecoin on Hedera. » Sinon : « It pays 1.0375 HBAR, a scaled demo amount. »]

9. Demo: Blocked — Mariem parle, Rihem lance les scénarios (50 s)

> « A spending system is judged by what it refuses. Budget: 100 dollars against a 4,150 dollar offer. Blocked, and the refusal itself is recorded on Hedera. Unverified supplier: in a 12-hour rush, the AI picks the cheapest broker; in our live run Claude even flagged a fraud risk, but the decision is not the AI's: Nasnes rejects, 3ases blocks. Duplicate payment: we replay the same payment. Refused, still one payment. Recycled invoice: even a verified supplier cannot bill twice. »

10. Vault: Hedera holds the money — Rihem, en direct (55 s)

[Scénario I (ACME) : Envoyer, montrer « Payment locked » et la transaction programmée sur HashScan, puis raison + PIN et Approve.]

> « Autonomy has a limit, and here Hedera enforces it. ACME needs 11,820 dollars of laptops, far above its 2,000 dollar limit. Above the limit, the money is not in the agents' account. It sits in a vault whose key is two keys: ours and a human approver's. Kanawita can only lock the payment: it creates a Hedera schedule that already carries its signature, one of two. Nothing can move. I approve with a reason and a PIN: the approver's signature is added, and Hedera itself executes the transfer. If I reject, the schedule is deleted. Even someone who steals our server's key cannot empty the vault. »

11. Guardian — Mariem (30 s)

> « Why is the guardian not an AI? Because you can argue with an AI, and someone will. 3ases is plain, deterministic code: budget, verified supplier, region, independent verification, invoice not reused, no previous payment, a sane price. One failed rule and the mission is blocked. Each organization has its own versioned policy, and twenty unit tests prove the same facts always give the same decision. »

12. Four Hedera services — Mariem (45 s)

> « How do we use Hedera? Four native services, no smart contract. The Consensus Service: every decision becomes a SHA-256 fingerprint, chained to the previous one and published, with no names, prices or documents on-chain. Scheduled transactions with a threshold key: the vault, where the human

> signature is enforced by the ledger. The Token Service: EJUSD, a demo stablecoin we created, so we pay the full amount. And the Mirror Node: anyone can re-check everything. »

13. Tampering and the public verifier — Mariem (45 s)

[Si possible en direct : Proof & verify, « Insider rewrites hashes too », puis « Open the public verifier ».]

> « So why not just a database? Because a database can rewrite its past. An insider edits the approved price: a normal check catches it. But a smart insider also recomputes every hash, or deletes the payment record. Locally, everything looks perfect. Hedera disagrees: TAMPERING DETECTED. And you don't have to trust our server either: our public verifier reads Hedera from your own browser and recomputes every fingerprint itself. Rihem? »

14. Verified live — Rihem (40 s)

> « This is not a mockup. Our eight original scenarios passed live on Hedera testnet with the real Claude. 114 automated tests cover the rules, the payments, the vault and the tamper detection. Next: real suppliers and USDC instead of our demo token, approvers signing from their own wallets, then two pilots in Tunisia, one NGO and one company. Who pays: any organization that lets agents spend, as a subscription plus a fee per proof; that is our hypothesis, not yet validated. And we say our limits openly: fictional suppliers, a demo token, and an approver key held by our server for the demo. »

15. Closing — Mariem puis Rihem (15 s)

> **Mariem :** « AI agents should act, but never outside rules nobody can argue with, never above the limit without a human signature, and never without proof anyone can check. »

> **Rihem :** « EL JOKER: autonomous, but never uncontrolled. The code is public and anyone can verify a mission. Thank you. Shoukran. »

[Restez sur la dernière slide pendant les questions : les liens GitHub, HashScan et /verify y sont affichés.]

Questions du jury

Répondez en 20 secondes maximum : une phrase de réponse, une preuve, et c'est tout. Si vous ne savez pas, dites « Good question, we haven't validated that yet » plutôt que d'inventer. Mariem répond à l'IA, au Guardian et à Hedera ; Rihem répond au positionnement, à la démo, au marché et à la suite.

Le positionnement

Isn't this just for NGOs? Emergencies are rare. (Rihem)
« No. The NGO is our first market and the hardest case. The engine is the same for a company: you saw ACME buying laptops under its own policy. Any organization that lets an agent spend needs limits, a human above the limit, and proof. »

Only one agent uses AI. Why is this an AI project? (Mariem)
« Because the hard problem is not calling an AI, it is letting it act on money safely. We use AI where it adds value, ranking offers and explaining trade-offs, and we wrap it in controls it cannot talk its way around. »

Why five agents instead of one? (Mariem)
« Separation of duties, like in a real organization: the one who chooses the supplier is not the one who approves or pays. Each agent has one job, and only one of them uses AI. »

Hedera

Why Hedera and not a normal database? (Mariem)
« A database can be rewritten by whoever controls it, including us. Hedera gives every decision an ordered, timestamped record that we cannot change, and it enforces the human signature on the vault. Slide 13 shows the attack only Hedera catches. »

How does the vault work, exactly? (Mariem)
« The vault account's key is a threshold key: two keys, ours and the approver's, both required. Kanawita creates a Hedera schedule that carries our signature only. When the approver signs, Hedera executes the transfer itself. If the operator rejects, we delete the schedule. »

Who holds the approver key? (Mariem)
« In the demo, our server holds it and uses it only after an authorized operator approves with a PIN. In production it lives in the approver's own wallet. On the ledger the guarantee is the same: our agents' key alone can never move the vault. »

What if the human approves after the schedule has expired? (Mariem)
« A Hedera schedule expires after about 30 minutes by default. Kanawita then locks a new schedule on the same payment, and it still needs both signatures. We have a test for it. »

Is EJUSD a real stablecoin? (Rihem)
« No. It is a demo token we created on testnet with Hedera's Token Service: fixed supply, no admin key. It lets us pay the full amount instead of a symbolic one. In production we would use a regulated stablecoin like USDC, which already exists on Hedera. »

Why not put all the data on-chain? (Mariem)
« Privacy. Beneficiary data, prices and invoices stay off-chain. On Hedera we only publish fingerprints, under 1 KB per message: enough to prove nothing changed, without exposing anyone. »

Why not a smart contract? (Mariem)
« We get what we need from native services: consensus, schedules with threshold keys, tokens and the Mirror Node. Our rules stay readable code that an auditor can test, and fees stay low and fixed. »

What if Hedera is down or slow? (Mariem)
« Then nobody gets paid. Kanawita waits until the Guardian's approval is anchored before paying. We retry the anchoring, and if a receipt is lost we ask again for the same transaction instead of sending a new one. »

Can I check a mission without trusting you? (Mariem)
« Yes. Open our public verifier: your browser downloads our claims, reads the proofs directly from the Hedera Mirror Node and recomputes every SHA-256 itself. If anything was edited or deleted, it shows TAMPERING DETECTED. »

Is this real or simulated? (Rihem)
« Real, on testnet: the payment you saw has a transaction ID on HashScan, and our topic 0.0.11007004 is public. The app also has a clearly labeled simulated mode for offline rehearsals. »
[Dites « the vault and EJUSD ran live » seulement après avoir lancé npm run demo:check avec le coffre configuré.]

L'IA et la sécurité

What if the AI hallucinates or gets prompt-injected? (Mariem)
« Then it recommends something wrong, and that's all it can do. Nasnes checks its claims against the registry, and 3ases blocks anything that breaks a rule. We even hide the registry status from the AI. »

Can the AI change the amount paid? (Mariem)
« No. The amount comes from the verified offer, and the payment is computed by code, never by the model. »

What happens if Claude is unavailable? (Mariem)
« After 45 seconds we switch to a simple rule-based ranking, clearly labeled FALLBACK. The rules and the payment checks don't change. »

Who writes the rules, and can someone change them quietly? (Mariem)
« Each organization has a versioned policy file. Its version includes a fingerprint of the file, and that version is written into every proof. If someone changes a rule, it shows on Hedera. »

How do you prevent paying twice? (Rihem)
« Three layers: an idempotency key per payment, database constraints that allow one payment per mission and per invoice, and a check that the invoice was never used. Replaying the payment returns an HTTP 409. »

Can an insider fake a human approval? (Rihem)
« The approval needs an authorized operator and a PIN checked on the server, and it is tied to a fingerprint of the exact facts reviewed. Above the limit, the money also needs the approver's signature on Hedera. In production we would add real identity. »

Marché et suite

Who are your customers, and who pays? (Rihem)
« Any organization that lets agents spend. We start with NGOs and grant programs, which must prove every dollar to donors, then companies' procurement. A subscription per organization plus a fee per proof. That is a hypothesis: we haven't validated it with customers yet. »

Have you talked to a real NGO or company? (Rihem)
« Not yet. The suppliers and both organizations in the demo are fictional. Our next step is two pilots in Tunisia, one NGO and one company. »
[Si vous avez parlé à quelqu'un d'ici le pitch, remplacez cette réponse par ce qui s'est vraiment passé.]

Who are your competitors? (Rihem)
« Agent frameworks and payment APIs give autonomy; audit tools give logs. We combine hard spending rules no AI can talk around, a human signature enforced by the ledger, and a public proof anyone can check. »

Does it scale? (Mariem)
« Each mission sends six to eight messages to Hedera, which is built for high throughput. The demo uses SQLite; production would use a server database. A new organization is just a new policy file. »

What would you do with three more months? (Rihem)
« Real supplier data, USDC instead of our demo token, approvers signing from their own wallets, and two pilots. »

What was the hardest part? (Rihem ou Mariem)
« A live testnet bug: a receipt timed out after the message was already accepted, so our retry created a duplicate proof and a false tampering alarm. We fixed it by asking for the receipt of the same transaction instead of resending, and added a test. »

Vidéo de démo (3 minutes)

Six moments : un paiement approuvé, une tentative de double paiement, un fournisseur piège, une entreprise au-dessus de sa limite (le coffre Hedera), une falsification détectée par le vérificateur public, et la conclusion. Enregistrez-la après avoir configuré le coffre et EJUSD, pour que tout soit réel.

Temps	À l'écran	Action	Voix off
0:00–0:12	Page d'accueil	Aucune	**Rihem :** « AI agents are starting to spend money. EL JOKER is the control layer: one AI proposes, rules decide, a human co-signs on Hedera, and Hedera proves everything. »
0:12–0:50	Conversation, scénario A	« Go for EL JOKER », Envoyer ; couper l'attente au montage	**Rihem :** « An NGO, 500 people in Gabès, 5,000 dollars. Claude picks the best offer, Nasnes checks ten facts, 3ases applies the NGO's rules: approved. Kanawita pays 4,150.00 EJUSD, our stablecoin on Hedera, and six decisions are anchored on Hedera. »
0:50–1:05	Panneau Actions	« Replay payment (attack) »	**Mariem :** « Someone sends the same payment again. Rejected. Still exactly one payment. »
1:05–1:25	Puce D	Envoyer, montrer BLOCKED	**Mariem :** « A trap: under time pressure, the AI picks an unverified broker. Nasnes rejects it, 3ases blocks it, nothing moves. »
1:25–2:10	Puce I (ACME), carte d'approbation	Envoyer ; montrer « Payment locked » et la transaction programmée sur HashScan ; raison + PIN, Approve	**Rihem :** « Same engine, a company: 11,820 dollars of laptops, above ACME's 2,000 dollar limit. Kanawita can't pay. It locks the payment in a Hedera vault: one of two signatures. I approve, the approver signs, and Hedera executes the transfer itself. »
2:10–2:50	Proof & verify, puis le vérificateur public	« Insider rewrites hashes too », puis « Open the public verifier »	**Mariem :** « An insider rewrites the price and every hash. Don't trust us: this page reads Hedera from your browser and recomputes everything. TAMPERING DETECTED. »
2:50–3:00	Slide finale	Aucune	**Les deux :** « EL JOKER: autonomous, but never uncontrolled. »

Le bouton de falsification reste grisé tant que toutes les preuves ne sont pas ancrées : attendez quelques secondes après le paiement.

Avant d'enregistrer

☐ Lancer `npm run hedera:setup -- --vault --stablecoin` depuis ton propre terminal, puis redémarrer l'application.
☐ Vérifier les badges : « AI · claude-opus-5-5 », « Hedera testnet », « Vault 2-of-2 », « Pays in EJUSD (HTS) ».
☐ Lancer `npm run demo:check` : toutes les lignes doivent être PASS.
☐ Vérifier qu'il reste des HBAR testnet sur le compte opérateur et des crédits sur le compte Anthropic.
☐ Faire une répétition complète sans enregistrer, pour connaître les temps d'attente.
☐ Navigateur en plein écran (1920×1080, zoom 110 %), mode Ne pas déranger de Windows activé.
☐ Ne jamais montrer le fichier .env, un terminal avec des clés, ni le portail Hedera.
☐ Enregistrer avec la Xbox Game Bar (Win + Alt + R) ou OBS, micro de casque, pièce calme.
☐ Au montage : couper les attentes, zoomer sur « 1 of 2 signatures » et sur TAMPERING DETECTED, ajouter des sous-titres en anglais.

☐ Exporter en MP4 1080p et vérifier la durée maximale imposée par le hackathon.